BoostFormer-Based Dual-View Awareness Feature Fusion Classification for Hyperspectral Images

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***Abstract*—Convolutional neural networks (CNNs) and Transformers have achieved significant progress in hyperspectral image (HSI) classification tasks. However, most existing methods tend to process spatial and spectral features independently, making it challenging to fully model the multilevel spatial–spectral correlations, which limits feature representation capabilities. In response to this issue, this article proposes a BoostFormer-based dual-view awareness feature fusion classification (BF-DFC) for HSIs to extract spatial–spectral joint features. Specifically, in the spatial view, the spatial feature awareness (SpaFA) module efficiently captures local details and global context, while the Spatial BoostFormer (SpaBF) establishes local–global hierarchical correlations to enhance the discriminative power of spatial features. In the spectral view, the spectral feature awareness (SpeFA) module extracts local and global interactive information from the spectral domain, and the Spectral BoostFormer (SpeBF) further models the interplay of local details and long-range dependencies to improve the robustness of spectral features. Finally, the co-enhanced dual-view feature fusion (CoDF) module performs fusion of spatial–spectral features through cross-view interaction and an adaptive weighting strategy, further enhancing the model's representation capabilities. Extensive experiments conducted on three hyperspectral datasets with diverse scenes and scales demonstrate that the proposed BF-DFC network consistently outperforms current state-of-the-art methods in classification accuracy with acceptable computational complexity. The source code will be available online at https://github.com/Yali-W/BFDFC**

***Index Terms*—Hyperspectral image (HSI) classification, information fusion, spatial–spectral features, Transformer, wavelet convolution.**

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I. Introduction

HYPERSPECTRAL images (HSIs), which contain abundant spatial and spectral information [1], [2], are widely applied in environmental monitoring [3], agricultural development [4], urban planning [5], and geological exploration [6]. Among applications, HSI classification is one of the most important techniques and has attracted widespread attention from researchers in recent years [7], [8], [9].

Traditional classification techniques include support vector machine (SVM) [10], principal component analysis (PCA) [11], random forest (RF) [12], and others. These traditional approaches typically extract features in a shallow manner, which may result in unstable performance when dealing with the high dimensionality and complexity of HSIs.

Recently, deep learning has driven significant progress in HSI classification [13], [14], [15]. Due to the good ability in local feature exploration, convolutional neural networks (CNNs) have been widely used in HSI classification. Initially, to explore the extraction of spectral and spatial information from HSIs, in the literature [16], the researchers introduced the spectral–spatial unified network (SSUN). This network simultaneously trains 1-D CNNs and 2-D CNNs to extract spectral and spatial features. In [17], recognizing the necessity for a joint model to analyze spectral and spatial information, the researchers proposed a novel 3-D CNN architecture. To further investigate the efficacy of CNNs in HSI classification, Ge et al. [18] developed a 2-D–3-D CNN and multibranch feature fusion network, which hierarchically extracts features from shallow to deep layers. Subsequently, considering the impact of multiscale features on land cover classification, Lee and Kwon [19] proposed a contextual deep CNN. This method constructs multiscale convolutional filters to model local spatial–spectral relationships between adjacent pixel vectors, capturing contextual interactions. Tu et al. [20] considered the spectral domain as a key component of contextual features and proposed a nonlocal dependency learning fully convolutional network (FCN) by combining global convolutional long short-term memory with context-aware module. To represent the spatial–spectral structure from multiple perspectives, Jia et al. [21] introduced a 3-D Gabor-CNN architecture that modulates CNN operations using multiscale and multidirectional 3-D Gabor filters. In order to reduce the number of convolutional kernel parameters, Zhao et al. [22] applied Gabor filters to

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modulate grouped separable convolutions. To improve the modeling capability of high-order spatial–spectral interactions and adapt to nonlinear variations, Zhang et al. [23] proposed a high-order deformable differential convolution network, which achieves explicit modeling of high-order feature interactions and adaptive capturing of spatial structural details. These methods have demonstrated the effectiveness of integrating spectral and spatial information through various network architectures.

Despite the remarkable success of CNN-based methods for HSI classification, several limitations persist. For instance, the inherent locality of convolutional operations restricts their receptive fields, leading to inadequate modeling of global contextual correlations. Although many researchers have employed multiscale convolutions to capture features with different kernel sizes [24], [25] or leveraged attention mechanisms to assist CNNs [26], [27], [28], it remains challenging to model the long-range dependencies between different regions of HSIs, especially in complex imaging environments. Recently, Transformer-based classification methods have gained significant attention. Unlike CNNs, the Transformer architecture captures long-range dependencies with its self-attention mechanism, which has led researchers to introduce Transformer models into the HSI classification domain [29], [30], [31]. He et al. [32] pioneered the HSI-BERT model, which employs self-attention mechanisms to model long-range pixel-wise dependencies. Hong et al. [33] revisited the challenges of HSI classification and introduced SpectralFormer, which partitions the spectral bands into multiple tokens to encode spectral information, thereby enabling long-range dependency modeling of spectral features. Subsequently, Zhong et al. [34] proposed a spectral–spatial Transformer network composed of a spatial attention module and a spectral correlation module. To enhance the discriminability of the center pixel in HSI classification, Feng et al. [35] proposed a center attention Transformer that generates stratified spatial–spectral tokens via superpixel sampling, addressing the issue of heterogeneous pixels introduced by patch sampling. Meanwhile, Yang et al. [36] proposed the center-focused Transformer, which achieves precise focusing on the center pixel through a Gaussian positional embedding and a cross-layer aggregation mechanism. Transformer-based HSI classification methods have made significant progress in modeling global long-range dependencies.

However, Transformer-based methods have limitations in extracting local fine-grained features. To address this critical issue, Shu et al. [37] have systematically explored CNN–Transformer hybrid architectures. Specifically, the spectral-guided multiscale feature-aware Transformer employs a multiscale feature-aware self-attention mechanism to dynamically integrate fine and coarse spatial features while enhancing spectral continuity through a spectral-guided fusion module. Furthermore, the dual feature aggregation network adopts a dual feature aggregation module to interactively aggregate local and global spectral–spatial features, effectively bolstering the capture of fine-grained information via multiscale neighborhood token aggregation [38]. Additionally, the spectral–spatial synergy guided attention network efficiently fuses local and global spectral–spatial features by stacking multiscale spectral extraction, a multibranch dual-shifting spatial enhancement module, and a guided token attention mechanism, thereby strengthening the extraction of global contextual information [39]. Wang et al. [40] proposed the mid-range convolutional modulated Transformer network. This method employs large-kernel convolution to simplify self-attention operations and introduces mid-range convolutional modulation to effectively capture mid-range spatial dependencies. To fully exploit local spatial–spectral joint information, Liu et al. [41] proposed a spatial–spectral multiscale tokenization Transformer and embedded a CNN–Transformer architecture into an unmixing module to obtain fine-grained subpixel details. Although this hybrid design partially compensates for the shortcomings of pure Transformers in capturing local details, CNNs naturally focus more on high-frequency information and inevitably lose spatial details during downsampling. To enlarge the receptive field while simultaneously modeling both high- and low-frequency information, inspired by the wavelet transform, this article introduces wavelet convolution [42], which is designed from both spatial and spectral dimensions. In the spatial dimension, 2-D wavelet convolution enlarges the receptive field while jointly modeling high- and low-frequency information. In the spectral dimension, 1-D wavelet convolution optimizes the interaction across channels.

Based on this, this article integrates the wavelet convolution with the Transformers proposing a BoostFormer-based dual-view awareness feature fusion classification (BF-DFC) for HSIs. Specifically, to enhance the model's perception of spatial and spectral features, we have designed the spatial feature awareness (SpaFA) and spectral feature awareness (SpeFA) modules. Additionally, to capture both local and global dependencies in the spatial and spectral domains, we propose the Spatial BoostFormer (SpaBF) and Spectral BoostFormer (SpeBF). Finally, to effectively fuse spatial and spectral information and further improve the model's performance, we introduce the co-enhanced dual-view feature fusion (CoDF) module. The main contributions of this article can be summarized as follows.

1) To enhance the awareness of spatial and spectral information, we design the SpaFA and SpeFA modules. These modules effectively represent and focus on the spatial and spectral information, thereby significantly improving the model's ability to perceive key information.
2) To enhance the network's ability to model local and global dependencies and improve its nonlinear representation capability, we combine multiscale deep convolution operations with a multihead self-attention mechanism to construct the SpaBF. Additionally, we designed the SpeBF by combining convolution operations with the self-attention mechanism.
3) To efficiently integrate spatial–spectral features, we propose the CoDF module, which aims to achieve efficient fusion of spatial and spectral features through the interaction of spatial and spectral features and a dynamic weight allocation strategy, ensuring the comprehensive and precise representation of features.
4) Extensive experiments are conducted on the PaviaU, HanChuan, and HongHu datasets. The results show that, compared to the current state-of-the-art methods based

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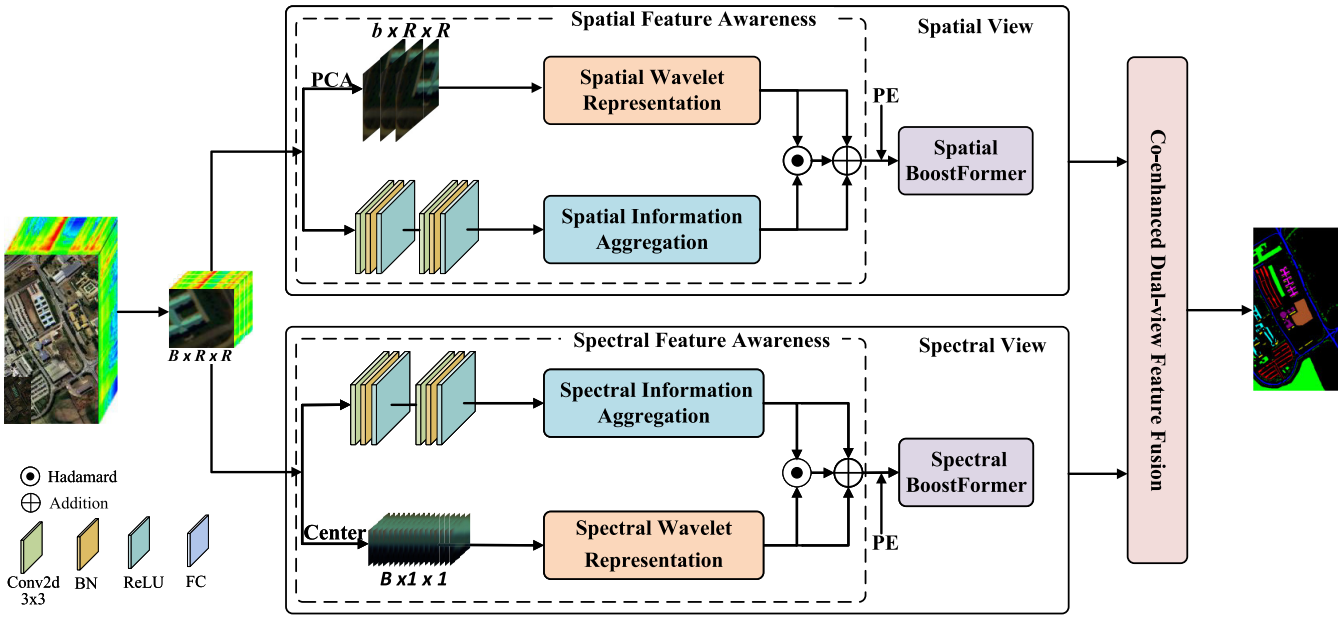


Fig. 1. Proposed BF-DFC classification network consists of spatial view, spectral view, and CoDF. The spatial view includes the SpaFA and SpaBF, while the spectral view contains SpeFA and SpeBF.

on CNN, Transformer, and Mamba, our BF-DFC model achieves stable and superior classification performance.

The remainder of this article is structured as follows. Section II offers the comprehensive description of the proposed BF-DAFC. Section III analyzes the experimental results in detail. Finally, Section IV concludes this article.

II. Methodology

This section provides an exploration of the proposed BF-DFC. First, we present an overview of the model's overall architecture. Subsequently, we delve into the design principles and implementation details of the key components.

A. Overview

An overview architecture of the BF-DFC, as shown in Fig. 1, contains three workflows: spatial view, spectral view, and CoDF view. Specifically, the original $X \in \mathbb{R}^{B\times H\times W}$ is divided into patches, denoted as $X_i \in \mathbb{R}^{B\times R\times R}$, where $R \times R$ represents the spatial size of the input patch and B denotes the number of spectral channels. In the spatial view, the inputs are X_i and X_i^{PCA}, where $X_i^{\text{PCA}} \in \mathbb{R}^{b\times R\times R}$ is obtained by applying PCA to the X_i sequence, and b represents the number of channels after PCA. In the spectral view, the inputs are X_i and X_i^{center}, where $X_i^{\text{center}} \in \mathbb{R}^{B\times 1\times 1}$ is the central pixel extracted from the X_i sequence. In the CoDF, the spatial view features and spectral view features are combined to generate a spatial–spectral joint representation.

B. Spatial View

To fully leverage spatial context information, we effectively extract spatial features from the spatial view. This view adopts a dual-module collaborative architecture, where the SpaFA module focuses on extracting multiscale spatial features, and the SpaBF structure establishes multilevel associations between local details and global context features, enhancing feature representation. This dual-module collaborative design not only effectively captures multiscale local spatial information but also models long-range spatial dependencies.

1) Spatial Feature Awareness: The SpaFA module consists of the spatial wavelet representation (SpaWR) module and the spatial information aggregation (SpaIA) module, as shown in Fig. 2. Specifically, the SpaWR module extracts multiscale spatial features with nonlinear mapping capabilities, while the SpaIA module focuses on key spatial information. The collaborative learning between these two modules significantly improves the model's ability to perceive spatial information.

a) Spatial wavelet representation: Standard convolution excels at encoding local features, but its fixed kernel design limits the model's ability to capture global context and multiscale frequency information. To address this, this article proposes SpaWR, which combines 2-D wavelet convolution (2DWTConv) with pointwise convolution (PWConv) in a cascaded manner, enabling effective extraction of both global context and multiscale frequency information.

2DWTConv decomposes the input features into multiscale frequency components, including low-frequency approximation components and high-frequency detail components, through wavelet transform, and performs convolution operations in parallel across each frequency band, expressed as

$$F_{wc}^{spa} = \text{IWT}\left(\text{Conv}_{3\times 3}\left(W, \text{WT}\left(X_i^{pca}\right)\right)\right) \tag{1}$$

where WT(·) and IWT(·) are wavelet Transformer and inverse wavelet transform, and $\text{Conv}_{3\times 3}(\cdot)$ denotes the convolution operation with a 3×3 kernel. W is the weight tensor of a $k \times k$ depth-wise kernel with four times as many input

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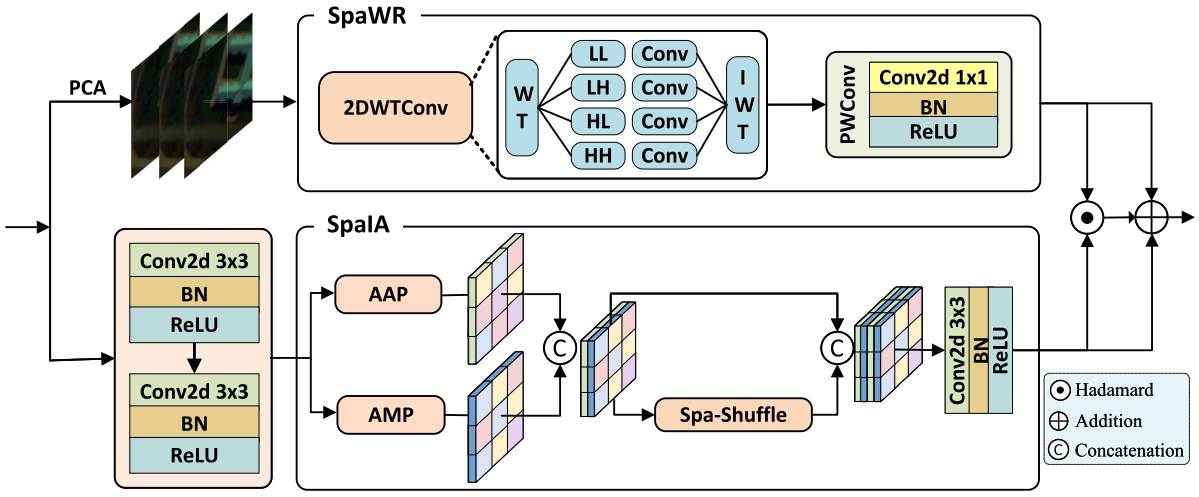


Fig. 2. Detailed structure of the SpaFA module, which includes SpaWR and SpaIA.

channels in X_i^{pca}. In this article, we use the 1-level Haar wavelet.

Then, the features extracted by WTConv are fed into PWConv to adjust the number of channels while learning a richer feature representation, which can be expressed as

$$F_{wr}^{spa} = \mathrm{PWC}\left(F_{wc}^{spa}\right) \tag{2}$$

where PWC(·) represents the PWConv layer (including convolution, batch normalization, and ReLU activation function).

b) Spatial information aggregation: To focus on spatial information, we propose the SpaIA module, which operates through three core stages: pooling, shuffling, and fusion.

1) Pooling: First, the hierarchical features are extracted through the convolution module, followed by the joint use of adaptive average pooling (AAP) and adaptive max pooling (AMP) to capture statistical features. Finally, the features are concatenated along the channel to generate a fused representation, as shown in the following:

$$F_{pool}^{spa} = \mathrm{Concat}\left(\mathrm{AAP}\left(\mathrm{Conv}\left(X_i\right)\right), \mathrm{AMP}\left(\mathrm{Conv}\left(X_i\right)\right)\right) \tag{3}$$

2) Shuffling: This stage introduces the stochastic permutation. Specifically, the pooled features are restored to a 2-D layout and then subjected to a random spatial shuffle, resulting in F_{shuf}^{spa}, to enhance robustness against spatial position perturbations;

3) Fusion: In this stage, the pooled features and the shuffled features are concatenated, and a convolution is applied to achieve cross-channel feature fusion

$$F_{ia}^{spa} = \mathrm{Conv}_{3\times3}\left(\mathrm{Concat}\left(F_{shuf}^{spa}, F_{pool}^{spa}\right), dim = c\right) \tag{4}$$

where "$dim = c$" means concatenation along the channel.

2) Spatial BoostFormer: To further enhance the network's ability to represent both spatial local and global features and improve its nonlinear expression capability, we introduce SpaBF. Unlike traditional Transformer modules, it employs multiscale multihead attention (MSMHA) to process local information, enabling the extraction of both multiscale local and global features.

As shown in Fig. 3, MSMHA is one of the key structures of SpaBF, aimed at enhancing the representational power of both local and global features. Specifically, the input T_{in} undergoes linear transformations to generate the query (Q), key (K), value (V), and projection (P). To effectively capture multiscale features, we partition V into three segments based on the number of attention heads, with each segment having the same number of attention heads. Subsequently, convolutional layers with varying kernel sizes are applied to each segment for feature extraction, thereby capturing multiscale information. Finally, all segment outputs are fused to obtain the multiscale feature representation V_m, calculated as follows:

$$V_m = \mathrm{Concat}\left(\mathrm{DWC}_{3\times3}\left(V\right), \mathrm{DWC}_{5\times5}\left(V\right), \mathrm{DWC}_{7\times7}\left(V\right)\right) \tag{5}$$

where $\mathrm{DWC}_{3\times3}(\cdot)$, $\mathrm{DWC}_{5\times5}(\cdot)$, and $\mathrm{DWC}_{7\times7}(\cdot)$ represent depthwise convolution operations with kernels of size 3×3, 5×5, and 7×7, respectively.

Then, the attention scores are calculated using the scaled dot-product operation by Q and K and weighted to V_m

$$\mathrm{MSA}\left(Q, K, V\right) = \mathrm{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right) V_m \tag{6}$$

where $(d_k)^{1/2}$ represents the contraction coefficient.

Finally, to enhance the model's dynamic adaptability to multiscale features and improve its nonlinear modeling capability, we introduce the feature modulator (P). The feature's adaptive calibration is achieved through the Hadamard product operation ($V_m \odot P$), allowing for dynamic adjustment based on the contribution of features at different scales, thereby optimizing feature representation

$$F_{bf}^{spa} = \mathrm{MSA}\left(Q, K, V\right) + V_m \odot P. \tag{7}$$

The forward propagation network is another key structure, where we use convolutional gated linear units (CGLUs) [43]. In CGLU, each token is associated with a unique gating signal, based on its closest fine-grained features, thereby enhancing the model's local modeling capability and robustness. As

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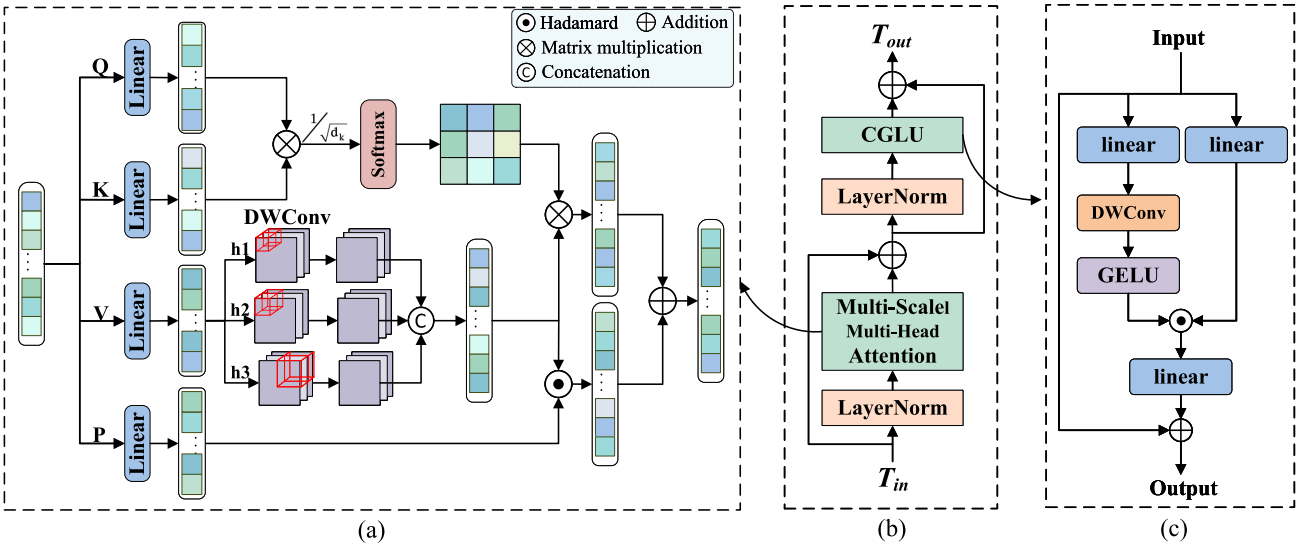


Fig. 3. Framework of SpaBF in (b). Structures of the MSMHA and CGLU are shown in (a) and (c), respectively.

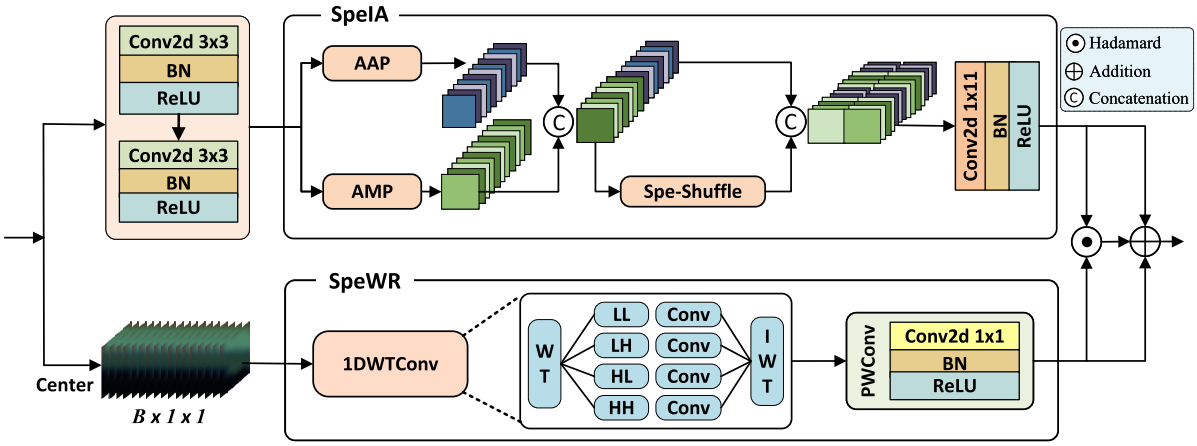


Fig. 4. Detailed structure of the SpeFA module, which includes SpeWR and SpeIA.

shown in Fig. 3(c), a simple 3 × 3 depthwise convolution is added before the activation function of the gated linear unit.

C. Spectral View

In response to the rich band composition and dense spectral information of HSI, we effectively extract spectral features from a spectral view. This view integrates the SpeFA and SpeBF structures. The SpeFA module efficiently captures spectral information, while the SpeBF structure collaboratively extracts local detailed spectral features and global spectral context information.

1) Spectral Feature Awareness: The SpeFA module consists of spectral wavelet representation (SpeWR) and spectral information aggregation (SpeIA), achieving precise and efficient extraction of key band information, as shown in Fig. 4. Specifically, SpeWR captures cross-band nonlocal correlations through multiscale wavelet decomposition, while SpeIA focuses on the important information of each band. The collaboration of these two components significantly enhances the model's ability to recognize spectral features.

a) Spectral wavelet representation: Considering the differences in feature expression across channels, where shallow channels often encode high-frequency details, while deep channels carry low-frequency semantics [44], we design the SpeWR module, which combines 1-D wavelet convolution (1DWTConv) with PWConv to achieve structured, multiscale channel mixing. Specifically, SpeWR leverages the multiresolution analysis property of the wavelet transform to establish interband correlations across multiple scales. In this module, we employ a 1DWTConv with a kernel size of 1 to decompose the channel vector at each spatial location along the channel axis, explicitly separating the channels into high- and low-frequency representations. Subsequently, pointwise convolution is applied to perform cross-channel feature reorganization and dimensionality transformation, which can be expressed as

$$F_{wr}^{spe} = \mathrm{PWC}\left(\mathrm{IWT}\left(\mathrm{Conv}_1\left(W, \mathrm{WT}\left(X_i^{center}\right)\right)\right)\right) \tag{8}$$

where $Conv_1$ represents a 1-D convolution with kernel 1.

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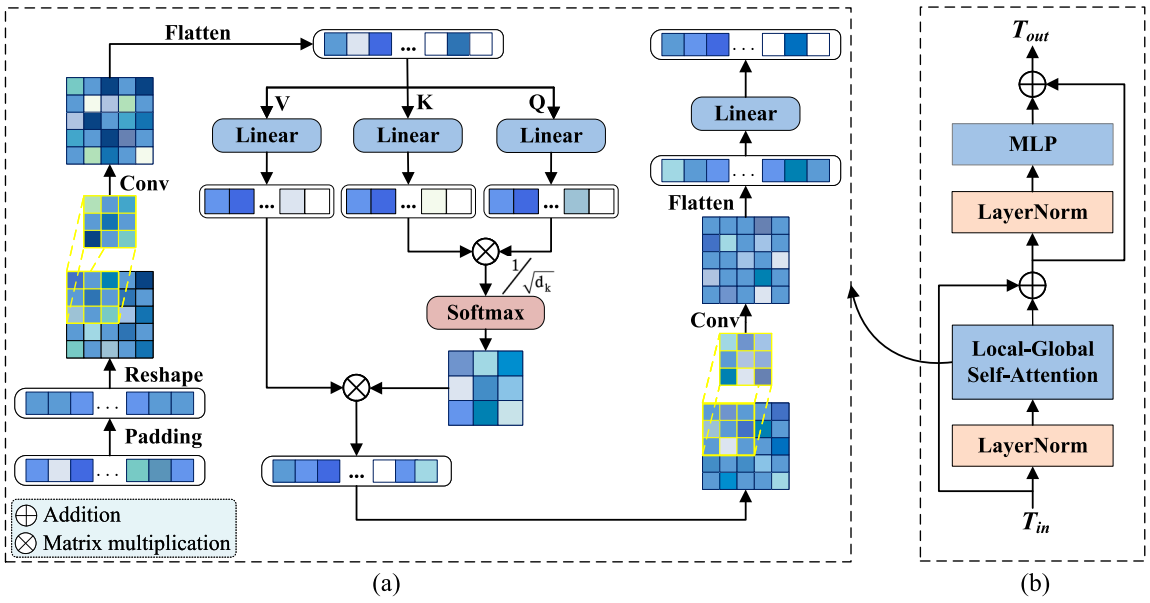


Fig. 5. Framework of SpeBF in (b). Structure of LGSA is shown in (a).

b) Spectral information aggregation: To effectively aggregate spectral information and enhance feature representation, we design a SpeIA module. This module first obtains the global values of the channels through pooling, then utilizes shuffling to achieve interband information interaction, and finally performs adaptive feature fusion to dynamically enhance key features. The specific formulation is as follows:

$$F_{pool}^{spe} = \text{Concat}\left(\text{AAP}\left(\text{Conv}\left(X_i\right)\right), \text{AMP}\left(\text{Conv}\left(X_i\right)\right)\right) \tag{9}$$

$$F_{ia}^{spe} = \text{Conv}_{1\times 11}\left(\text{Concat}\left(F_{shuf}^{spe}, F_{pool}^{spe}\right), dim = w\right). \tag{10}$$

In (9), Conv(·) represents the convolution block operation. In (10), $\text{Conv}_{1\times 11}(\cdot)$ indicates a 1×11 convolution operation, and "$dim = w$" indicates that the concatenation is along the width. F_{shuf}^{spe} represents channel shuffle features of the concatenated features.

2) Spectral BoostFormer: In the spectral view, to fully exploit local and nonlocal information and enhance the network's representation capability, we propose the SpeBF. As shown in Fig. 5, local–global self-attention (LGSA) is one of the key components of SpeBF. LGSA has two components: one part uses 2-D convolution to extract local information, and the other is the self-attention module for modeling the global spectral information.

Specifically, the spectral features extracted by SpeFA are first flattened into $F_{s\times c}^{spe}$. Then, $F_{s\times c}^{spe}$ is padded and transformed into a 2-D spectral feature tensor

$$F_{s_1\times s_1}^{spe} = \text{Reshape}\left(F_{s\times c}^{spe}, (pad_1, pad_2)\right) \tag{11}$$

where pad_1 and pad_2 denote the padded features in the last dimension, specifically pad_1 units on the left side and pad_2 units on the right side, with the replication of edge values for padding. c represents the number of channels and s is the product of width and height. Then, the feature $F_{s_1\times s_1}^{spe}$ is processed through convolutional layers and a self-attention module to obtain local and global spectral information

$$F_{bf}^{spe} = \text{Conv}_{\text{k}\times\text{k}}\left(\text{SA}\left(\text{conv}_{k\times k}\left(F_{s_1\times s_1}^{spe}\right)\right)\right) \tag{12}$$

where $\text{Conv}_{\text{k}\times\text{k}}(\cdot)$ represents the convolutional layer operation (with k = 3) and SA(·) represents the self-attention operation.

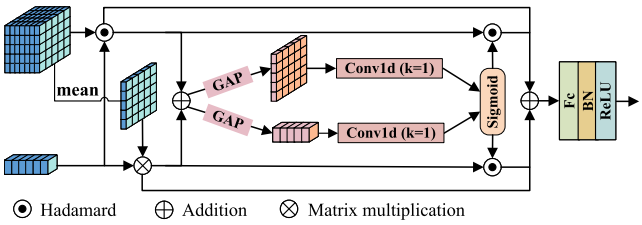


Fig. 6. Co-enhanced dual-view feature fusion.

D. Co-Enhanced Dual-View Feature Fusion

To achieve more effective integration of spatial features and spectral features, we propose the CoDF module. As illustrated in Fig. 6, the spectral view feature F_{bf}^{spe} first undergoes the element-wise multiplication with spatial view features F_{bf}^{spa}, producing an enhanced spatial representation

$$F_{co}^{1} = F_{bf}^{spa} \odot F_{bf}^{spe} \tag{13}$$

where $\odot$ denotes the Hadamard product. Simultaneously, the channel-wise mean of the spatial view features F_{bf}^{spa} is calculated and integrated into the spectral view through matrix multiplication to generate an augmented spectral representation

$$F_{co}^{2} = \left[\phi_{c=\text{mean}}\left(F_{bf}^{spa}\right)\right] \otimes F_{bf}^{spe} \tag{14}$$

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where $\otimes$ represents matrix multiplication and $\phi_{c=\text{mean}}(\cdot)$ denotes the channel-wise mean. Then, the spatial–spectral features are fused through addition

$$F'_{fusion} = F^1_{co} \oplus F^2_{co}. \tag{15}$$

After the initial fusion, we applied global average pooling (GAP), 1-D convolution, and sigmoid to obtain two attention weights for adaptive weighted fusion of the spatial–spectral features

$$F^{spa}_{att} = \text{Sigmoid}\left(\text{Conv}\left(\text{GAP}\left(F'_{fusion}\right)\right)\right) \odot F^1_{co} \tag{16}$$

$$F^{spe}_{att} = \text{Sigmoid}\left(\text{Conv}\left(\text{GAP}\left(F'_{fusion}\right)\right)\right) \odot F^2_{co} \tag{17}$$

$$F''_{fusion} = F^{spa}_{att} \oplus F^{spe}_{att} \oplus F'_{fusion}. \tag{18}$$

The fused features are then passed to the classifier to obtain the estimated classification probabilities P for the corresponding pixels. The classification loss is computed by comparing the predicted probabilities with the ground truth labels. In this study, we adopt the cross-entropy loss as the selected loss function, as shown in the following equation:

$$CEloss = -\frac{1}{N}\sum_{i=1}^{N}\sum_{j=1}^{M}\left(y_{ij}\log\left(p_{ij}\right)\right) \tag{19}$$

where N and M represent the number of samples and the number of categories, respectively; y_{ij} represents the true label; and $p_{ij} \in P$.

III. Experimental and Analysis

In this section, we first introduce the dataset, followed by the experimental setup, which includes configurations, evaluation metrics, comparison methods, and hyperparameter settings. Next, we provide a comprehensive comparison of the state-of-the-art methods from both qualitative and quantitative perspectives. Finally, we conduct detailed ablation experiments to analyze the effectiveness of the proposed modules.

A. Datasets

To comprehensively evaluate the effectiveness of the proposed method, we selected three diverse HSI datasets: Pavia University (PU), WHU-Hi-HanChuan (HC), and WHU-Hi-HongHu (HH).

1) Pavia University: The PU dataset was acquired in 2003 using the ROSIS-03 over the urban area surrounding the University of Pavia, Italy. The spatial size is 610 $\times$ 610 pixels with a ground sampling distance of 1.3 m/pixel. The image contains 115 spectral bands spanning the wavelength range of 0.43–0.86 μm, with 12 bands discarded due to noise contamination, resulting in 103 retained spectral channels. The dataset comprises 42 371 annotated pixels spanning nine land cover classes.

2) WHU-Hi-HanChuan: The HC dataset was acquired on June 17, 2016 over Hanchuan City, Hubei Province, China, using an airborne Headwall Nano-Hyperspec sensor with a 17-mm focal length. The hyperspectral cube exhibits dimensions of 1217 $\times$ 303 pixels, containing 274 spectral bands spanning 400–1000 nm. The dataset incorporates 256 810 annotated pixels distributed across 16 land cover classes.

TABLE I
Number of Training, Validation, and Testing Samples for Each Class on the PU Dataset

ID	Color	Category	Train.	Val.	Test.
1		Asphalt	30	15	6586
2		Meadows	30	15	18604
3		Gravel	30	15	2054
4		Trees	30	15	3019
5		PaiMetShe	30	15	1300
6		BareSoil	30	15	4984
7		Bitumen	30	15	1285
8		SelfBloBri	30	15	3637
9		Shadows	30	15	902
		Total	270	135	42371

TABLE II
Number of Training, Validation, and Testing Samples for Each Class on the HC Dataset

ID	Color	Category	Train.	Val.	Test.
1		Strawberry	30	15	44690
2		Cowpea	30	15	22708
3		Soybean	30	15	10242
4		Sorghum	30	15	5308
5		Water spinach	30	15	1155
6		Watermelon	30	15	4488
7		Greens	30	15	5858
8		Trees	30	15	17933
9		Grass	30	15	9424
10		Red roof	30	15	10471
11		Gray roof	30	15	16866
12		Plastic	30	15	3634
13		Bare soil	30	15	9071
14		Road	30	15	18515
15		Bright object	30	15	1091
16		Water	30	15	75356
		Total	480	240	256810

3) WHU-Hi-HongHu: The HH dataset was acquired on November 20, 2017, over Honghu City, Hubei Province, China. The dataset encompasses a diverse range of crop varieties, including multiple varieties within a single crop species. Data collection employed a Headwall Nano-Hyperspec imaging sensor with 17-mm focal length mounted on a DJI Matrice 600 Pro UAV platform. The HSI with 940 $\times$ 475 pixel dimensions containing 270 spectral bands across 400–1000 nm. The airborne HSI demonstrates a spatial resolution of approximately 4.3 cm.

Tables I–III present the land cover class names, corresponding false colors for each land cover category, and the detailed information on the number of training, validation, and test samples for three datasets.

B. Experimental Setup

1) Configuration: All experiments were conducted on the same hardware, featuring an Nvidia GeForce RTX 3090 GPU with 24 GB of GPU memory. The proposed BF-DFC was implemented using Python 3.8.19 and the PyTorch 2.4.1 framework. Adam was used as the optimizer, with a batch size of 64, a learning rate set to 0.0005, and 200 training epochs. The input patch size for the network is set to 23, and the PCA dimensionality reduction is set to 12. No data augmentation

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TABLE III
NUMBER OF TRAINING, VALIDATION, AND TESTING SAMPLES FOR EACH CLASS ON THE HH DATASET

ID	Color	Category	Train.	Val.	Test.
1		Red roof	30	15	13996
2		Road	30	15	3467
3		Bare soil	30	15	21776
4		Cotton	30	15	163240
5		Cotton firewood	30	15	6173
6		Rape	30	15	44512
7		Chinese cabbage	30	15	24058
8		Pakchoi	30	15	4009
9		Cabbage	30	15	10774
10		Tuber mustard	30	15	12349
11		Brassica parachinensis	30	15	10970
12		Brassica chinensis	30	15	8909
13		Small Brassica chinensis	30	15	22472
14		Lactuca sativa	30	15	7311
15		Celtuce	30	15	957
16		Film covered lettuce	30	15	7217
17		Romaine lettuce	30	15	2965
18		Carrot	30	15	3172
19		White radish	30	15	8667
20		Garlic sprout	30	15	3441
21		Broad bean	30	15	1283
22		Tree	30	15	3995
		Total	660	330	385713

was applied to the method, and the other method-specific hyperparameters are followed the original optimal settings. In each experiment, the training and validation sets are composed of 30 and 15 samples randomly selected from the image, respectively. The test set consisted of the remaining samples.

2) Evaluation Metrics: In our experimental evaluation, we employed three metrics for HSI classification: overall accuracy (OA), average accuracy (AA), and Kappa. To ensure statistical robustness and mitigate potential biases from random sample selection, we conducted comprehensive comparative experiments and systematic ablation studies. Each experimental configuration was executed ten times with ten random seeds, and we recorded both the mean accuracy values and their corresponding standard deviations to provide a reliable assessment of model performance.

C. Comparison With State-of-the-Art Methods

To comprehensively evaluate the effectiveness of the proposed method, several state-of-the-art deep learning techniques are selected as comparison methods, including CNN-, Transformer-, and Mamba-based approaches. The detailed descriptions of the comparison methods are provided as follows.

1) For CNN-Based:

1) *CVSSN [45]:* This network is a central vector-oriented self-similarity network that uses adaptive weight addition and Euclidean distance to capture central-neighbor spectral–spatial relationships.
2) *CLOLN [46]:* This lightweight spectral–spatial network reduces CNN parameters and computational complexity via intralayer channel-oriented feature extraction and interlayer neighbor-pixel-aware activation.
3) *SSEFN [47]:* This network performs information enhancement from both spatial and spectral dimensions to improve feature discrimination, thereby achieving efficient feature learning under few-sample conditions.
4) *CASANet [48]:* The network employs a cascaded spatial cross-attention mechanism to integrate local and global spatial features and propagates key regional information across channels through a group cascade structure, thereby enhancing network robustness.

2) For Transformer-Based:

1) *MASSFormer [49]:* This method converts spectral–spatial features into memory tokens that store prior knowledge via a memory tokenizer, and introduces these tokens into self-attention using a memory-augmented Transformer encoder, thereby alleviating information loss and improving HSI classification performance.
2) *GSC-Vit [50]:* This model utilizes a grouped separable convolution and vision Transformer to extract both local and global spectral–spatial information.
3) *DSFormer [51]:* The model incorporates a kernel selection Transformer block, which dynamically selects the appropriate kernel size to optimize the receptive field for effective spatial–spectral feature extraction.
4) *ClusterFormer [52]:* This method employs dual-branch deformable convolution and spatial–spectral cluster attention to balance local and global information in a hierarchical Transformer, significantly improving classification performance.
5) *LRDTN [53]:* This method integrates dynamic convolution, multiscale enhancement, and a local–global Transformer to effectively capture local multiscale context and long-range dependencies in HSIs while reducing model parameters and improving classification performance.

3) For Mamba-Based:

1) *MambaHSI [54]:* A Mamba-based image-level method that models long-range interactions across the entire image via spatial–spectral Mamba blocks, adaptively fusing features for efficient pixel-wise classification.
2) *HSIMFormer [55]:* This method combines an inner scale Transformer for short-range modeling and a cross-scale Mamba for long-range modeling, fully exploiting multiscale spatial–spectral information, thereby significantly improving HSI classification performance.

1) Quantitative Results and Analysis: The quantitative results of the different methods are presented in Tables IV–VI, where the bold font indicates the best performance and the underlined entries denote the second-best results. As shown in Table IV, BF-DFC achieves the best performance on the PU dataset, obtaining the highest OA, AA, and Kappa coefficient among all competing methods. These results demonstrate that the proposed framework can effectively exploit spectral–spatial information and generate discriminative feature representations for urban scenes with relatively

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TABLE IV
CLASSIFICATION RESULTS (ACC% ± STD%) ON THE PU DATASET

Class	CVSSN [2023]	CLOLN [2024]	SSEFN [2025]	CSCANet [2025]	MASSFormer [2024]	GSC-ViT [2024]	DSFormer [2025]	ClusterFormer [2025]	LRDTN [2025]	MambaHSI [2024]	HSIMFormer [2025]	BF-DFC [our]
1	**99.03±0.59**	93.84±2.24	98.61±0.99	96.84±1.40	93.38±4.61	98.16±1.09	93.70±1.60	89.28±5.67	98.37±1.34	93.91±1.44	91.94±3.73	98.48±1.26
2	98.79±0.69	99.32±0.54	98.54±0.78	86.69±0.71	96.45±2.46	97.32±1.91	97.81±0.58	84.37±6.03	**99.50±0.23**	95.62±2.83	93.91±2.62	97.70±2.54
3	87.64±7.68	93.00±4.92	70.65±6.74	92.63±3.21	93.02±2.78	87.13±8.14	74.74±6.81	79.76±7.98	97.64±2.54	93.15±5.46	93.36±2.65	**98.68±1.96**
4	**98.00±1.13**	69.06±14.13	92.18±7.74	95.65±1.69	97.15±1.13	95.05±2.88	84.85±5.50	84.73±8.62	87.53±8.00	90.31±2.84	93.55±2.47	97.23±1.64
5	99.72±0.55	86.39±0.18	**100.00±0.00**	99.76±0.10	99.76±0.24	99.77±0.28	99.15±0.47	99.95±0.08	99.53±0.29	**100.00±0.00**	99.67±0.37	99.99±0.02
6	85.94±5.34	88.50±8.03	90.14±7.26	**99.97±0.04**	96.68±3.17	86.64±7.98	80.35±7.04	98.36±1.90	94.98±3.24	98.64±0.77	98.60±1.83	99.55±0.77
7	86.04±6.16	81.98±10.00	73.67±5.82	99.91±0.10	99.95±0.10	82.59±9.29	81.45±5.82	97.21±5.58	89.53±7.84	96.74±2.64	99.82±0.19	**100.00±0.00**
8	91.34±1.99	91.36±2.24	85.75±2.14	93.94±2.14	92.38±4.76	89.48±6.12	81.15±4.45	98.14±1.96	89.99±4.76	92.11±5.60	92.25±3.58	**98.14±1.20**
9	99.08±0.92	87.25±5.45	94.47±7.98	98.55±0.56	98.80±0.51	99.33±0.84	73.64±7.40	91.17±8.36	92.81±5.38	99.39±0.73	97.26±1.29	**99.87±0.17**
OA(%)	95.29±0.86	91.51±3.05	92.87±1.30	92.44±0.45	95.79±1.48	93.97±1.53	89.58±1.59	88.78±2.09	96.25±1.19	95.16±1.75	94.39±1.61	**98.81±0.72**
AA(%)	93.95±1.11	87.85±2.91	89.33±1.89	95.99±0.43	96.40±0.83	92.83±1.91	85.21±1.76	91.44±1.80	94.43±1.86	95.54±1.33	95.60±0.80	**98.85±0.32**
Kappa	93.79±1.11	88.93±3.83	90.62±1.63	90.20±0.57	94.45±1.90	92.04±2.01	86.39±1.98	85.59±2.55	95.05±1.56	95.08±2.34	92.65±2.07	**98.42±0.94**

TABLE V
CLASSIFICATION RESULTS (ACC% ± STD%) ON THE HC DATASET

Class	CVSSN [2023]	CLOLN [2024]	SSEFN [2025]	CSCANet [2025]	MASSFormer [2024]	GSC-ViT [2024]	DSFormer [2025]	ClusterFormer [2025]	LRDTN [2025]	MambaHSI [2024]	HSIMFormer [2025]	BF-DFC [our]
1	93.59±1.46	92.34±1.96	95.09±0.59	89.35±2.66	91.11±3.48	94.94±0.88	94.98±0.71	80.23±15.66	**98.41±0.56**	91.37±5.67	92.40±3.01	95.45±2.41
2	94.45±1.42	76.11±26.83	95.38±0.85	87.89±2.56	87.03±2.03	92.95±1.94	92.60±2.31	92.08±3.99	**97.20±1.23**	81.02±7.38	84.87±2.39	90.75±2.44
3	76.21±4.32	68.21±10.95	75.61±4.64	90.10±1.52	92.40±3.29	79.68±5.52	76.29±3.67	63.76±25.76	92.11±4.13	94.34±2.81	91.79±3.55	**97.45±3.03**
4	90.85±3.44	84.25±7.40	83.20±2.32	98.53±0.15	98.74±0.89	95.48±3.82	92.33±5.17	91.87±8.52	95.60±5.34	98.42±1.18	98.25±1.41	**99.33±0.48**
5	56.64±8.44	20.4±8.94	31.47±4.27	99.91±0.14	99.71±0.39	50.74±9.97	36.30±5.60	99.38±0.80	81.31±9.59	99.65±0.73	99.19±0.92	**99.98±0.03**
6	45.27±5.47	31.9±19.29	56.18±6.73	79.42±6.27	82.23±4.53	45.19±10.28	52.21±5.23	72.45±6.67	66.87±6.21	77.59±5.98	86.42±3.56	**92.12±2.54**
7	75.75±4.34	40.93±11.68	47.61±2.91	**96.63±0.62**	93.23±2.51	81.55±7.95	64.19±2.81	75.70±16.29	78.94±8.58	92.47±5.19	95.34±1.68	96.29±4.24
8	89.49±4.36	74.56±10.65	89.08±2.04	76.05±2.00	79.49±3.69	91.74±2.82	80.8±3.95	81.72±6.82	**94.45±3.58**	77.19±6.30	79.36±3.59	87.87±4.58
9	75.52±4.94	70.35±18.28	71.93±3.75	83.33±2.84	86.89±3.23	69.14±4.63	58.96±14.49	72.54±6.48	72.54±6.48	88.63±2.89	86.57±3.82	**93.46±3.91**
10	96.29±2.06	82.03±15.54	85.60±4.90	97.68±0.79	97.95±1.42	88.81±12.44	92.04±3.34	83.76±14.55	95.07±1.94	93.01±3.37	97.41±0.72	**98.79±0.39**
11	**96.13±3.87**	76.82±9.57	58.13±8.30	93.58±2.16	94.77±3.08	93.64±3.79	73.02±3.11	70.73±19.24	88.76±3.90	86.63±7.63	95.62±1.96	95.66±3.70
12	77.59±6.95	42.20±23.01	43.30±6.84	98.43±0.85	95.79±2.46	55.56±6.87	40.72±4.05	96.05±2.94	90.36±6.18	92.96±4.83	95.97±2.24	**99.36±0.76**
13	65.30±5.65	63.54±20.03	53.07±6.83	72.82±3.49	79.62±3.43	60.31±8.53	49.13±4.61	62.80±17.64	76.82±7.65	79.30±3.54	76.92±4.60	**85.26±7.39**
14	87.87±3.86	79.27±12.15	83.75±3.57	87.86±3.09	88.52±7.65	84.72±3.73	79.60±5.09	43.72±17.91	**95.05±2.07**	90.04±2.81	83.82± 4.31	93.30±2.90
15	86.52±4.27	59.67±14.35	55.38±12.14	**97.19±0.97**	96.08±4.76	75.98±12.97	57.43±15.07	96.04±4.09	74.23±14.14	94.11±1.92	93.36±3.80	93.37±3.30
16	97.05±1.61	98.31±1.75	99.69±0.19	97.45±0.91	98.23±1.19	99.33±0.63	99.31±0.35	84.47±11.06	**99.92±0.18**	98.12±1.90	98.07±1.24	98.55±1.21
OA(%)	88.82±1.07	77.29±4.57	81.99±1.43	90.79±0.61	91.75±0.77	87.95±1.72	84.17±0.96	77.95±4.91	93.25±0.65	90.70±1.93	91.67±0.74	**95.76±0.61**
AA(%)	81.53±0.90	66.31±4.14	70.28±0.84	90.39±0.38	91.12±0.65	79.01±1.98	71.92±1.58	78.76±3.24	87.35±1.25	89.68±1.37	90.96±0.60	**94.81±0.93**
Kappa	86.92±1.26	73.76±5.00	79.18±1.57	89.28±0.70	90.39±0.89	85.97±1.97	81.62±1.10	74.75±5.35	92.14±0.75	89.16±2.59	90.29±0.85	**95.04±0.71**

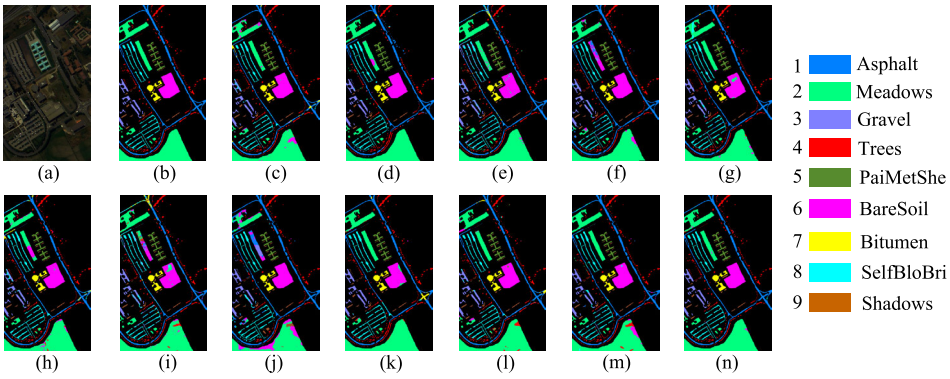


Fig. 7. Qualitative visualization of the classification map on the PU dataset. (a) False-color image. (b) Ground-truth map. (c) CVSSN. (d) CLOLN. (e) SSEFN. (f) CSCANet. (g) MASSFormer. (h) GSC-ViT. (i) DSFormer. (j) ClusterFormer. (k) LRDTN. (l) MambaHSI. (m) HSIMFormer. (n) BF-DFC.

homogeneous structures. Table V presents the quantitative results on the HC dataset. BF-DFC consistently outperforms the compared approaches across all evaluation metrics. Since the HC dataset contains more complex land-cover distributions and richer spatial details, the superior performance indicates that the proposed method is capable of capturing complementary spectral–spatial dependencies and distinguishing categories with high interclass similarity. The classification

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TABLE VI
CLASSIFICATION RESULTS (ACC% ± STD%) ON THE HH DATASET

Class	CVSSN [2023]	CLOLN [2024]	SSEFN [2025]	CSCANet [2025]	MASSFormer [2024]	GSC-ViT [2024]	DSFormer [2025]	ClusterFormer [2025]	LRDTN [2025]	MambaHSI [2024]	HSIMFormer [2025]	BF-DFC [our]
1	98.47±0.47	97.77±1.06	98.07±0.51	94.32±1.04	97.48±0.61	94.85±3.19	96.80±1.29	87.35±11.74	**98.56±0.85**	96.56±1.21	94.68±2.47	95.79±2.15
2	83.16±3.89	74.17±6.46	73.24±5.33	**98.29±0.66**	94.89±3.21	79.56±3.96	68.99±2.45	87.85±10.62	67.53±7.14	95.79±2.16	92.41±2.98	96.83±3.43
3	**98.20±0.72**	94.61±4.42	90.62±2.21	86.55±0.91	88.29±2.93	97.55±1.91	94.47±1.37	65.91±20.81	98.16±1.09	94.59±1.11	87.34±2.85	92.00±3.92
4	99.53±0.16	99.05±0.53	99.20±0.15	92.71±1.77	95.47±1.31	99.68±0.10	99.22±0.19	78.95±19.82	**99.78±0.15**	97.63±1.68	95.33±2.00	98.06±0.67
5	54.48±13.33	38.76±9.74	25.94±3.65	96.14±1.10	92.76±2.36	51.82±13.91	47.68±8.20	85.81±10.79	74.92±5.43	96.69±1.72	96.12±2.72	**98.82±1.03**
6	97.01±1.08	93.43±3.00	91.66±1.94	92.06±0.39	94.72±2.21	96.42±0.70	94.90±1.25	91.75±5.39	**98.56±0.78**	99.06±2.49	93.25±1.48	94.47±2.11
7	92.95±1.82	89.40±4.50	84.90±2.02	80.13±4.02	82.09±5.57	91.26±4.57	90.42±3.09	75.62±12.38	**93.31±2.70**	86.57±3.31	82.42±3.31	87.33±4.06
8	53.71±8.59	44.67±11.20	47.54±5.90	90.01±2.09	90.55±2.68	42.93±5.53	41.12±4.45	90.58±14.77	51.07±7.01	95.77±3.08	91.32±2.72	**96.09±3.02**
9	**99.39±0.35**	97.81±1.65	96.63±1.12	94.12±0.36	97.09±1.92	98.97±0.57	98.55±0.35	94.89±2.63	97.77±2.13	94.97±1.50	98.02±1.44	97.61±2.26
10	88.90±2.43	81.09±5.30	87.87±2.60	86.07±1.37	87.87±1.03	89.24±3.79	87.08±2.90	91.42±5.74	**96.13±1.42**	89.73±2.96	88.10±3.58	93.65±2.52
11	78.04±4.14	83.27±14.18	80.79±4.46	87.13±1.85	87.65±3.33	77.42±7.13	81.67±4.74	89.61±8.46	79.32±11.03	91.46±2.87	90.84±2.69	**92.13±2.81**
12	71.36±6.46	62.19±10.68	57.06±5.74	80.28±2.50	83.31±4.79	69.31±5.43	65.26±2.81	70.13±11.30	82.18±8.07	87.23±4.30	84.68±3.66	**91.17±4.17**
13	82.01±4.78	80.39±6.47	80.53±2.26	73.39±3.94	78.86±3.51	84.40±3.56	73.71±3.36	70.34±10.96	84.36±4.73	**88.90±3.87**	78.25±3.50	88.56±4.39
14	90.75±2.95	88.33±7.31	78.49±3.43	96.00±1.63	95.65±1.07	88.18±4.36	76.45±3.03	85.30±13.24	92.31±1.90	93.69±1.94	94.37±1.76	**97.47±1.79**
15	82.85±5.37	44.55±18.37	38.37±4.29	**99.94±0.08**	99.24±0.50	74.55±16.27	79.64±4.88	95.03±3.45	70.98±7.25	98.79±1.04	98.02±1.94	99.57±0.77
16	97.99±0.87	97.17±1.8	93.94±3.15	95.63±0.71	97.13±0.66	**98.48±0.89**	98.07±0.79	97.92±2.20	96.90±2.20	97.64±1.26	94.58±2.27	98.16±0.94
17	88.15±3.95	81.89±6.14	71.73±5.54	99.17±0.32	98.12±1.41	89.44±4.57	87.83±2.04	78.33±21.92	93.54±4.14	98.00±2.24	95.51±6.60	**99.77±0.25**
18	67.10±5.84	45.11±17.36	62.01±2.65	**99.71±0.19**	99.12±0.47	75.19±6.44	75.95±4.32	96.17±6.27	78.11±6.53	96.98±1.99	98.15±1.15	98.90±0.86
19	84.70±4.63	79.05±8.14	75.11±3.36	94.64±1.14	**95.80±1.51**	89.45±3.67	85.38±2.69	82.56±10.14	85.14±5.41	91.56±2.87	93.57±1.61	93.61±2.23
20	83.36±3.98	68.78±10.05	67.58±6.15	98.02±1.11	96.72±1.68	79.18±4.63	73.84±5.53	96.64±2.63	82.06±10.59	**99.19±0.43**	96.54±2.88	99.04±0.82
21	32.27±5.58	25.12±7.16	44.80±4.58	**100.00±0.00**	99.07±1.60	32.87±5.93	31.73±3.30	97.42±3.86	43.85±12.04	99.86±0.40	99.94±0.11	**100.00±0.00**
22	51.64±11.24	49.56±5.69	36.68±4.24	99.20±0.70	98.96±1.08	54.67±11.52	58.55±7.34	95.53±5.46	75.59±10.68	99.51±0.55	98.02±1.45	**99.70±0.47**
OA(%)	90.75±1.18	86.59±1.38	84.21±1.24	90.32±0.75	92.51±0.81	90.34±1.08	88.89±0.80	81.67±9.12	93.04±0.62	94.88±0.90	92.61±0.73	**95.85±0.60**
AA(%)	80.73±1.35	73.46±1.64	71.94±0.78	92.43±0.20	93.10±0.44	79.93±1.21	77.61±0.95	86.60±3.31	83.64±1.00	94.79±0.78	92.55±0.42	**95.85±0.43**
Kappa	88.45±1.42	83.28±1.67	80.54±1.40	87.97±0.88	90.63±0.98	87.97±1.29	86.10±0.95	78.27±9.68	91.25±0.77	94.35±1.35	90.73±0.89	**94.78±0.76**

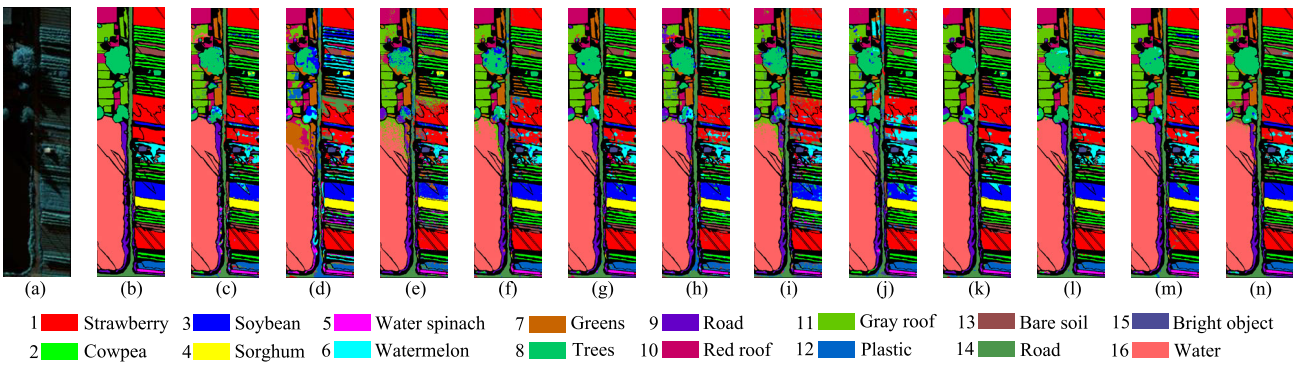


Fig. 8. Qualitative visualization of the classification map on the HC dataset. (a) False-color image. (b) Ground-truth map. (c) CVSSN. (d) CLOLN. (e) SSEFN. (f) CSCANet. (g) MASSFormer. (h) GSC-ViT. (i) DSFormer. (j) ClusterFormer. (k) LRDTN. (l) MambaHSI. (m) HSIMFormer. (n) BF-DFC.

results on the HH dataset are summarized in Table VI. BF-DFC again achieves the highest OA, AA, and Kappa coefficient among all competing methods. Owing to the larger spatial coverage and more complicated agricultural environments of the HH dataset, these results further demonstrate the effectiveness and robustness of the proposed framework in handling large-scale hyperspectral scenes with diverse class distributions. Overall, BF-DFC consistently achieves superior performance on all three datasets, highlighting its capability to learn discriminative spectral–spatial representations and its robustness under diverse hyperspectral scenarios.

2) Visual Evaluation and Analysis: Figs. 7–9 illustrate the classification maps of the proposed method and the comparative methods for all three datasets. From the visualization results in Fig. 7, it can be seen that most of the comparative methods exhibit significant misclassifications between Class 2 (Grassland) and Class 6 (Bare Soil). This category confusion is mainly due to the high spectral similarity between these two land cover types. In contrast, the proposed BF-DFC method optimizes the spatial–spectral feature representation through a spatial and spectral feature-aware mechanism, significantly enhancing the interclass discriminability between grassland and bare soil, thereby effectively alleviating this classification challenge. Fig. 8 shows that most of the comparative algorithms perform poorly on the Trees and Soybean categories, with noticeable misclassifications. The BF-DFC method, however, achieves higher recognition accuracy for these two categories, with more complete category representation and clearer boundaries in the classification results. Fig. 9 shows that most comparative algorithms tend to misidentify parts of the Cotton as Cotton Firewood. In comparison, the BF-DFC method exhibits a lower misclassification rate, and

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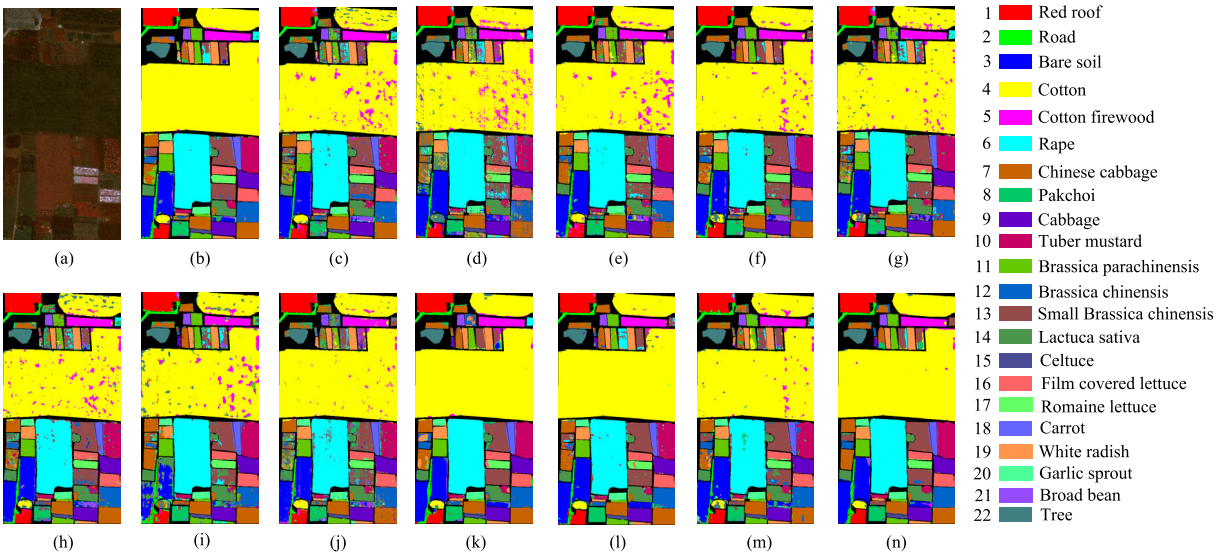


Fig. 9. Qualitative visualization of the classification map on the HH dataset. (a) False-color image. (b) Ground-truth map. (c) CVSSN. (d) CLOLN. (e) SSEFN. (f) CSCANet. (g) MASSFormer. (h) GSC-ViT. (i) DSFormer. (j) ClusterFormer. (k) LRDTN. (l) MambaHSI. (m) HSIMFormer. (n) BF-DFC.

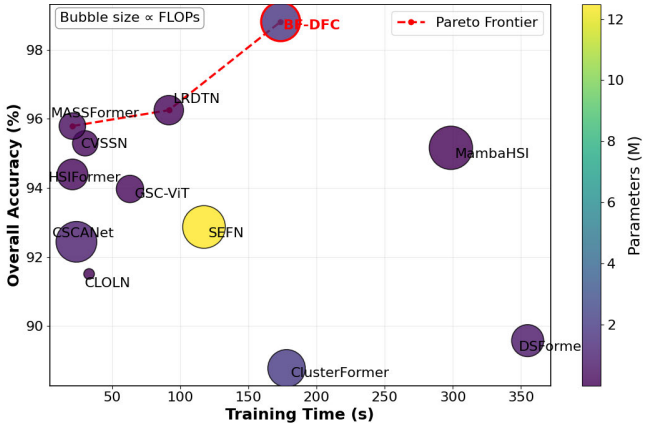


Fig. 10. Pareto frontier analysis of different methods on the PU dataset in terms of OA, average training time, FLOPs, and parameter quantity. Bubble size denotes the FLOPs, while lighter colors indicate larger parameter quantities.

TABLE VII

COMPLEXITY ANALYSIS OF DIFFERENT METHODS ON THE PU DATASET. FLOPS DENOTE THE COMPUTATIONAL COST PER BATCH OF HSI SAMPLES. TRAIN/TEST TIME INDICATES TRAINING AND INFERENCE TIME UNDER A PATCH-BASED SETTING

Metric	params/M	FLOPs/G	Train time/s	Test time/s
CVSSN	0.247	0.020	30.23	4.75
ClOLN	0.005	0.002	33.09	7.28
SEFN	12.481	3.370	117.40	9.86
CASANet	0.848	1.698	23.78	5.75
MASSFormer	0.304	0.026	20.82	3.05
GSC-ViT	0.153	0.032	63.10	3.88
DSFormer	0.677	0.108	354.97	119.35
ClusterFormer	1.967	0.557	178.01	4.74
LRDTN	0.137	0.053	91.60	10.86
MambaHSI	0.043	4.381	298.62	0.18
HSIMFormer	0.171	0.074	20.90	10.08
BF-DFC	1.786	0.887	173.64	19.87

the classification results for each category are more complete, further confirming its superior classification performance.

3) Model Complexity Analysis: To compare different methods from the perspective of model efficiency, Table VII reports the runtime, model complexity, and computational cost of each method on the PU dataset, while Fig. 10 further illustrates the tradeoff between classification performance and computational efficiency. The proposed BF-DFC achieves the highest OA of 98.81%, consistently outperforming all competing methods. As shown in Fig. 10, BF-DFC lies on the Pareto frontier, indicating that it achieves a superior balance between accuracy and efficiency. Although BF-DFC does not have the lowest training time, it delivers significantly higher classification accuracy than both faster models (e.g., CVSSN and CLOLN) and computationally heavier models (e.g., MambaHSI and DSFormer). These results demonstrate that BF-DFC achieves state-of-the-art performance while maintaining competitive computational efficiency.

D. Ablation Study

1) Ablation Study of the CoDF Module: To validate the effectiveness of the model, we designed a series of ablation experiments, including the spatial view, spectral view, and dual-view fusion experiments. The fusion experiments include the CoDF module, as well as addition and concatenation experiments. The experimental results are presented in Table VIII. The experimental results show that the hyperspectral joint model outperforms the single view. Moreover, the proposed CoDF fusion module outperforms direct addition or concatenation (after averaging to the same dimension) in terms of classification accuracy.

2) Ablation Experiments for the SpaBF, SpaFA, SpaWR, and SpaIA Modules: To validate the effectiveness of the SpaBF, SpaIA, and SpaWR modules, we designed four sets of comparative experiments. The experimental results are

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TABLE VIII
CLASSIFICATION RESULTS (ACC% ± STD%) WITH DIFFERENT FEATURE FUSION PROCESSES OF THE SPATIAL VIEW AND SPECTRAL VIEW

DT	Spatial View	Spectral View	Add	Concat	CoDF	OA(%)	AA(%)	Kappa(%)
PU	✓					95.03±1.16	92.93±1.39	93.44±1.52
		✓				92.59±2.79	95.89±1.01	90.37±3.52
	✓	✓	✓			96.55±1.71	96.71±0.38	95.47±2.22
	✓	✓		✓		97.77±1.23	98.16±0.12	97.05±1.60
	✓	✓			✓	**98.81±0.72**	**98.85±0.32**	**98.42±0.94**
HC	✓					93.62±0.40	93.37±0.36	92.57±0.48
		✓				89.36±1.57	87.35±2.27	87.62±1.81
	✓	✓	✓			92.98±1.83	87.94±2.57	91.82±2.10
	✓	✓		✓		92.05±1.73	92.59±0.14	90.75±1.98
	✓	✓			✓	**95.76±0.61**	**94.81±0.93**	**95.04±0.71**
HH	✓					94.50±0.68	94.47±0.44	93.08±0.84
		✓				92.16±1.01	91.11±0.82	90.15±1.25
	✓	✓	✓			94.48±0.56	94.35±0.15	93.06±0.71
	✓	✓		✓		94.18±0.88	94.12±0.64	92.68±1.10
	✓	✓			✓	**95.85±0.60**	**95.85±0.43**	**94.78±0.76**

TABLE IX
CLASSIFICATION RESULTS (ACC% ± STD%) OF THE SPATIAL VIEW

DT	Module	OA(%)	AA(%)	Kappa(%)
PU	Spa-Transformer	82.12±2.22	81.70±2.89	76.77±2.83
	SpaBF	91.61±2.77	90.49±0.71	89.10±3.52
	SpaBF+SpaIA	92.17±1.97	91.19±0.51	89.76±2.45
	SpaBF+SpaFA+SpaWR	**95.03±1.16**	**92.93±1.39**	**93.44±1.52**
HC	Spa-Transformer	70.56±3.03	67.32±1.36	66.32±3.11
	SpaBF	91.56±0.59	90.87±0.38	90.17±0.67
	SpaBF+SpaIA	92.72±0.58	90.88±1.39	91.50±0.68
	SpaBF+SpaFA+SpaWR	**93.62±0.42**	**93.37±0.36**	**92.57±0.48**
HH	Spa-Transformer	78.27±2.69	75.46±2.77	73.26±3.06
	SpaBF	93.49±0.70	93.06±0.87	92.23±0.39
	SpaBF+SpaIA	93.86±0.60	93.70±0.22	92.28±0.75
	SpaBF+SpaFA+SpaWR	**94.50±0.68**	**94.47±0.44**	**93.08±0.84**

TABLE X
CLASSIFICATION RESULTS (ACC% ± STD%) OF THE SPECTRAL VIEW

DT	Module	OA(%)	AA(%)	Kappa(%)
PU	Spe-Transformer	73.18±2.23	71.30±1.47	66.17±2.50
	SpeBF	76.11±3.96	81.55±2.57	69.75±4.53
	SpeBF+SpeIA	88.12±3.09	90.57±2.24	84.69±3.72
	SpeBF+SpeFA+SpeWR	**92.59±2.79**	**95.89±1.01**	**90.37±3.52**
HC	Spe-Transformer	69.74±1.23	66.30±1.12	65.31±1.33
	SpeBF	72.99±1.18	65.03±0.82	68.80±1.31
	SpeBF+SpeIA	90.15±0.93	88.30±0.40	88.52±1.08
	SpeBF+SpeFA+SpeWR	**89.36±1.57**	**87.35±2.27**	**87.62±1.81**
HH	Spe-Transformer	65.63±2.62	62.50±1.15	59.36±2.56
	SpeBF	68.71±1.89	65.79±1.58	62.68±2.01
	SpeBF+SpeIA	91.80±0.43	91.63±0.56	89.71±0.53
	SpeBF+SpeFA+SpeWR	**92.16±1.01**	**91.11±0.82**	**90.15±1.25**

shown in Table IX. The first set of experiments used a traditional Transformer model for spatial feature extraction, which yielded relatively poor classification performance. The second set of experiments incorporated the SpaBF module, effectively addressing the shortcomings of traditional Transformers in modeling local information and significantly improving classification performance. The third set of experiments further integrated the SpaIA module with SpaBF, resulting in a marked improvement in classification performance compared to the second set, thereby validating the effectiveness of the SpaIA module. The fourth set of experiments incorporated the SpaWR module on top of SpaBF and SpaIA, further improving classification performance compared to the third set of experiments, thereby demonstrating the effectiveness of the SpaWR module. Additionally, the final experimental results indicate that the combination of the SpaBF and SpaFA modules produces a significant synergistic effect.

3) Ablation Study of the SpeBF, SpeFA, SpeWR, and SpeIA Modules: We designed four sets of experiments to evaluate the contributions of the SpeBF, SpeFA, SpeWR, and SpeIA modules. The results are shown in Table X. The experimental results indicate that the classification performance of the second set of experiments is significantly better than that of the first set across all datasets, validating the effectiveness of the proposed SpeBF module. Furthermore, the performance of the third set of experiments has been further improved compared to the second set, demonstrating that the SpeIA module can effectively enhance feature extraction capabilities. In comparison to the third set of experiments, the fourth set achieved performance improvements on all datasets except for the HC dataset, where the classification results slightly decreased due to dense sample distribution. This result also proves the effectiveness of the SpeWR module.

Based on the comprehensive results of the ablation experiments, it is demonstrated that each proposed module is both innovative and effective. Furthermore, the proposed BF-DFC network enables efficient integration of spatial and spectral information, enhancing the classification accuracy.

E. Parameter Analysis

1) Patch Size: To investigate the impact of patch size on classification performance, we maintained other parameters

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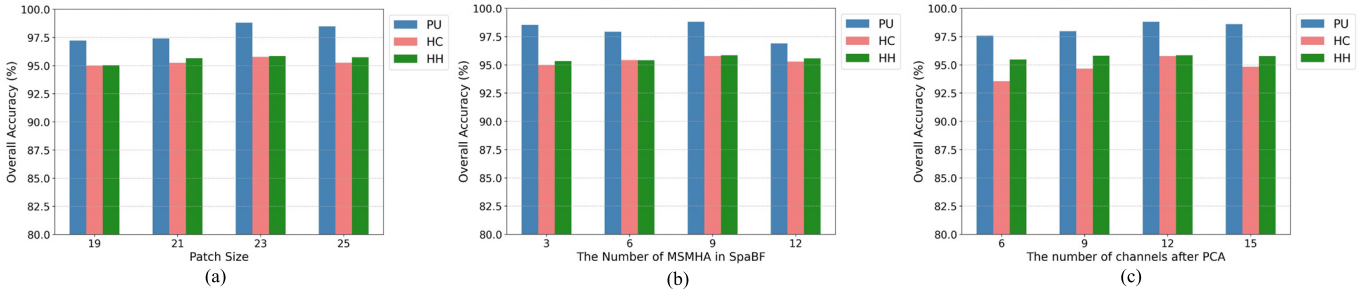


Fig. 11. Impact of different parameters on OA. (a) Patch size. (b) Number of MSMHA in SpaBF. (c) Number of channels after PCA.

constant and considered a set of candidate values 19, 21, 23, and 25 for the patch size. Fig. 11(a) depicts the model's OA across various patch sizes for the three datasets. The experimental results show that on these three datasets, the classification performance remains generally stable across different patch sizes. When the patch size is set to 23, the classification accuracy on all three datasets reaches a relatively optimal level; therefore, we set the patch size to 23.

2) Number of MSMHA in SpaBF: The number of MSMHA is a critical parameter in SpaBF. We considered a set of candidate values: 3, 6, 9, and 12. Fig. 11(b) illustrates the impact of different MSMHA quantities on OA. From the figure, we observe that when MSMHA is set to 9, the OA results across all three datasets are optimal. Therefore, we set the number of MSMHA to 9.

3) Number of Channels After PCA: Regarding the selection of the number of channels after PCA dimensionality reduction, we examined the impact of reducing the dimensionality to 6, 9, 12, and 15 on classification performance, while keeping other parameters constant, to analyze the effect of spectral dimension reduction on model performance. As shown in Fig. 11(c), the experimental results demonstrate that the classification performance on the three datasets is optimal when the dimensionality is reduced to 12 via PCA; therefore, we set the PCA dimensionality reduction to 12.

IV. Conclusion

This study proposes a novel network architecture, BF-DFC, designed to effectively capture and fuse the complex spatial and spectral information of HSIs, thereby improving classification accuracy. Specifically, the SpaFA module focuses on enhancing spatial feature perception, while the SpeFA module excavates discriminative spectral information. Subsequently, the SpaBF module effectively captures both local and global spatial representations, improving the nonlinear representation capability of spatial features. Meanwhile, the SpeBF module models both short-range and long-range spectral correlations to strengthen spectral feature representation. Finally, the CoDF fusion module integrates spatial and spectral information to further enhance overall feature representation capability. Extensive experiments conducted on three benchmark hyperspectral datasets demonstrate that BF-DFC achieves superior classification performance compared with existing methods. Although the proposed method introduces moderately increased computational cost compared with some lightweight models, it provides a favorable tradeoff between classification accuracy and computational efficiency.

Our study still has several limitations that deserve further investigation. We observed that the classification accuracy exhibits slight fluctuations as the number of wavelet convolution layers increases. Although these fluctuations are relatively limited, future work will explore adaptive modeling strategies based on dynamic wavelet bases to further enhance the stability and adaptability of the proposed model.

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